

Attribute	iOS	iPadOS	macOS	watchOS	visionOS
UI framework	SwiftUI, UIKit	SwiftUI, UIKit	SwiftUI, AppKit	SwiftUI, WatchKit (WKApplicationDelegateAdaptor)	SwiftUI, RealityKit
Scene types	scene-based life cycle; WindowGroup	UIWindowScene / UISceneSession; WindowGroup(id:for:)	WindowGroup, Window, UtilityWindow, Settings, MenuBarExtra	single-target watch app: dependent or independent watch app	WindowGroup, volume, ImmersiveSpace
Primary input	touch	touch, pointer (UIPointerInteraction), Apple Pencil	pointer, keyboard, trackpad	Digital Crown, double tap	look and tap (gaze privacy, hover effects)
Windowing model	one scene per app instance	multitasking: resizable windows via UIApplicationSupportsMultipleScenes, menu bar	window sizing, placement, style, level and restoration per scene; menu bar and menu bar extra	TabView with verticalPage, NavigationStack (Digital Crown moves between full-screen pages)	Shared Space (windows, volumes) and Full Space (ImmersiveSpace, mixed/progressive/full)
Widget / complication surfaces	Home Screen, Lock Screen, StandBy, CarPlay, Notification Center	Home Screen, Today view, StandBy, CarPlay, Notification Center	Mac desktop and Notification Center	complications: watch faces and Smart Stack (accessoryCircular, accessoryRectangular, accessoryCorner)	accessory widgets on the visionOS wall (systemLarge, systemExtraLargePortrait)
Background model	BackgroundTasks: BGAppRefreshTask, BGProcessingTask; UIBackgroundModes	BackgroundTasks: BGAppRefreshTask, BGProcessingTask	App Nap; BackgroundTasks framework is not on macOS at all	WKExtendedRuntimeSession; background refresh (WKApplicationRefreshBackgroundTask); WatchConnectivity background tasks	BackgroundTask.appRefresh(:) exists on iOS, iPadOS, visionOS and watchOS
Distribution channel	App Store (App Store Connect, TestFlight)	App Store (App Store Connect, TestFlight)	Mac App Store, or Developer ID (direct distribution) signed and notarized	App Store from the watch (independent) or with the iPhone app (dependent)	App Store: compatible "Designed for iPad" apps offered automatically, or a native visionOS app
Min Xcode / SDK	Xcode 27, 27 SDK, deployment minimum 26	Xcode 27, 27 SDK, deployment minimum 26	Xcode 27; Apple silicon only (Rosetta is not for this)	Xcode 27, 27 SDK, deployment minimum 26	Xcode 27, 27 SDK, deployment minimum 26

CONDITIONAL COMPILATION & AVAILABILITY

Pattern	Evaluated	FieldNotes example
<code>#if os(macOS)</code>	Compile time	Settings and MenuBarExtra scenes; denser NoteRow padding on Mac
<code>if #available(iOS 27, macOS 27, watchOS 27, visionOS 27, *)</code>	Run time	the on-device summary button (new in 27)
<code>if #unavailable(iOS 27, ...)</code>	Run time	one-time pin migration on 26
<code>@available(iOS 27, ..., *)</code>	Compile time (enforced at call sites)	SummaryCard; callers must be guarded
<code>@available(watchOS, unavailable)</code>	Compile time	exportNotebook(:) on Apple Watch

NOTES

- `os()` takes `iOS`, `macOS`, `watchOS`, `tvOS`, `visionOS` or `Linux` — there is no `iPadOS` condition, since iPad builds use the iOS SDK.
- A false `#if` branch is never type-checked for that platform, so it may call APIs that don't exist there.
- The trailing `*` in `#available(iOS 27, *)` means "on every other platform, from its minimum deployment".